

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1393

Wheeler 138 kV
May 17, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1393. The Interconnection Customer proposes a 83.6 MW solar photovoltaic facility interconnecting at the Wheeler 138 kV substation in the Virginia area. The request was received on January 22, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 468 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,786,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WHEELER – PASCO CNTY SOLID WASTE RESOURCE RECOVERY 138 kV	THOMSTON – LOUDON 345 kV line	110.6	510	49.3
WOODS DP – TOA VACA 138 kV #1	WOODS DP – TOA VACA 138 kV #2	104.3	475	18.3
WOODS DP – TOA VACA SUBSTATION 138 kV line	WOODS DP – TOA VACA 138 kV Circuit 1	104.3	475	18.3
THOMASTON – COMANCHE PEAK 345 kV	THOMASTON – HELIJET 345 kV	102.3	1,319	8.8
PASCO CNTY SOLID WASTE RESOURCE RECOVERY – VMEA PEAKING GEN 138 kV Line	NOVEC PARADISE – PASCO CNTY SOLID WASTE RESOURCE RECOVERY 138 kV line	101.0	314	43.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WHEELER – PASCO CNTY SOLID WASTE RESOURCE RECOVERY 138 kV Line	Rebuild line with higher-capacity conductor	\$4,410,000	16% (\$707,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WOODS DP – TOA VACA 138 kV #1	Reconductor line	\$629,000	11% (\$71,000)
WOODS DP – TOA VACA 138 kV #2	Reconductor line	\$363,000	11% (\$41,000)
THOMASTON – COMANCHE PEAK 345 kV	Reconductor line	\$25,921,000	2% (\$647,000)
PASCO CNTY SOLID WASTE RESOURCE RECOVERY – VMEA PEAKING GEN 138 kV line	Reconductor line	\$1,163,000	28% (\$320,000)
	Total allocated to project		\$1,786,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.